

ASHLEY LIU

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EDUCATION

University of California, San Diego - M.S., Computer Engineering (EC79) Expected 12/2026
Coursework: *Algorithms, Computer Architecture, Machine Learning, Web Mining & Recommender Systems, NLP, Computer Vision, Deep Generative Models, AI Agents*

Lancaster University & Beijing Jiaotong University - Dual B.S./B.E. in Computer Science, Major GPA 3.76 09/2020 - 06/2024
Coursework: *Operating Systems, Computer Networks, Distributed Systems, Data Structures & Algorithms, Compilers, Security, AI*

WORK EXPERIENCE

Software Engineer | AI/ML Intern, California ISO (CAISO), Folsom, CA 06/2026 - 09/2026

- Owned end-to-end development of CTRACC, a production NERC PER-005 training and qualification platform built with JavaScript, PHP, SQL Server, and IIS/FastCGI, integrating task, course, evaluation, and qualification workflows.
- Designed a version-aware relational model separating stable task identity from revision-specific records; preserved exact task-version links across qualification cards, course objectives, learner assignments, requalification, and point-in-time reporting.
- Implemented an 8-stage deterministic retrieval pipeline over 120+ approved, versioned articles using field-weighted BM25F, four-lane Reciprocal Rank Fusion, concept/alignment scoring, confidence gating, and query/candidate score logging.
- Hardened the shared IIS/FastCGI runtime through SQL integrity checks, load testing, session optimization, worker tuning, and automated deployment checks; verified task-version, completion, and reporting consistency across modules.

Software Development Engineer, Advanced Micro Devices, Foxborough, MA 04/2025 - 10/2025

- Built and deployed an internal APU/GPU lifecycle-management platform using Spring Boot, REST APIs, MyBatis, MySQL, and Vue 3, modeling asset/request state transitions with RBAC, validation, audit traceability, and ticket/waitlist workflows through NGINX and CI/CD.
- Extended KitChecker and QuickEPR across C#/VB.NET codebases; built C++/Python automation for test execution, result validation, and failure triage across Linux and Windows environments.
- Debugged next-generation GPU failures across functional correctness, driver compatibility, system stability, and performance, working across firmware/driver boundaries with Linux, DirectX, Vulkan, OpenGL, GPU PerfStudio, Radeon GPU Analyzer.

Technical Consultant, AX Future Academy, Boston, MA 12/2024 - 04/2025

- Built and maintained Java/Spring backend services and data-driven web features for an education platform, implementing search/ranking workflows and supporting cloud-hosted deployment and platform maintenance.

Image Processing Engineer Intern, Zuoyouming Technology Co., Ltd, Hubei, CN 07/2023 - 09/2023

- Built reusable PyTorch U-Net and YOLOv5 computer-vision pipelines for 4,340 UAV orthophotos, covering preprocessing, augmentation, annotation conversion, dataset validation, training, post-processing, and batch inference for scalable geospatial feature extraction.

Android Remote Control Application Developer Intern, Hikeen Technology Co., Ltd, Shenzhen, CN 07/2022 - 09/2022

- Designed and implemented a modular Android TV control stack with separate UI, networking, and command-processing layers, using UDP broadcast discovery, session management, sequence/ACK/retry logic, and asynchronous sockets to maintain reliable low-latency command delivery and gesture translation under unstable network conditions.

PROJECT EXPERIENCE

BJTU University Online Library Upgrades - Backend Engineering, Information Center 09/2023 - 02/2024

- Designed a Spring Boot backend and relational data model for an online library platform, implementing JSON APIs for catalog search, pagination, categories, sorting, and frontend integration with Vue 3 and MyBatis/MySQL.
- Implemented authentication and authorization with Spring Security, JWT, RBAC, and bcrypt; integrated NGINX reverse proxy/load balancing for service routing and deployment.

Agentic Chart Reasoning System | Multi-Agent AI / Multimodal Reasoning 03/2026 - 05/2026

- Built a Python multi-agent reasoning system on the OpenAI API with modular routing, specialized solvers, critics, and verification agents.
- Designed evidence-grounded verification and targeted repair to detect inconsistent outputs and recover from agent failures, improving GPT-4o-mini strict exact-match accuracy on DVQA hard reasoning tasks from 36.7% to 70.0%.

PUBLICATIONS AND SELECTED AWARDS

- **Liu, X.; Tao, H.** "Java-Based Development of a Vehicle Speed Recommendation Model for Assisted Driving." IEEE Conference on Machine Learning, Algorithms and Artificial Intelligence (MLAAI), 2023. (Paper ID: MLAAI-2368)
- **1st Prize**-National Univ. Intelligent Vehicle Competition (Vision); Outstanding Graduation Project (Multimodal Lip Recognition, 2023 - 2024); Academic Scholarship (2022 - 2023); Outstanding Student (2020 - 2021); Global Leadership Award (10/2022)

SKILLS

Languages: Java, C++, Python, TypeScript/JavaScript, SQL, C#, Go, Bash/Shell

Backend / Full-Stack: Spring Boot, Spring Security, REST APIs, Node.js, Flask, React, Vue 3, MyBatis, JPA, JWT, RBAC, MySQL, SQL Server

Systems / Infrastructure: Linux, Windows, Docker, NGINX, IIS/FastCGI, HTTP, TCP/UDP, Distributed Systems, Git, CI/CD, Maven/Gradle

AI/ML: PyTorch, TensorFlow, Transformers, OpenCV, RAG, LLM Applications, Integration Testing, Test Automation, Performance Profiling